END-TO-END RETAIL PERFORMANCE & BEHAVIORAL ANALYTICS

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ABSTRACT

Retail organizations generate vast volumes of transactional and customer data, yet converting this data into actionable business insights remains a challenge. This project presents an end-to-end retail analytics solution aimed at evaluating sales performance, customer behaviour, product profitability, store efficiency, and return patterns.

The analysis is conducted using cleaned transactional data and follows a structured workflow involving data preprocessing, exploratory analysis, relational database design, SQL-based business analysis, and interactive visualization using Power BI. Key performance indicators such as revenue, profit, return rate, and customer value were derived and analysed across multiple dimensions including time, region, category, and sales channel.

The final output is a five-page interactive dashboard that enables stakeholders to monitor business performance, identify high-value customers and products, assess store efficiency, and understand return behaviour. The insights derived from this analysis support data-driven decision-making and provide recommendations for improving profitability, customer engagement, and operational efficiency.

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INTRODUCTION

Company Name: RetailNova Inc.

Industry: Retail & E-Commerce

Founded: 2016

Headquarters: Chicago, USA

Sales Channels: Online platform, 60+ physical stores across 8 regions.

RetailNova Inc. is a mid-sized retail and e-commerce company operating across multiple regions with a combination of physical stores and online sales channels. The company offers a diverse product portfolio including electronics, apparel, home & kitchen, and personal care products.

With increasing competition and evolving customer expectations, RetailNova faces challenges in understanding sales performance trends, identifying high-value customers, managing product returns, and optimizing store operations. Retail analytics plays a crucial role in addressing these challenges by enabling organizations to convert raw transactional data into actionable insights.

This project focuses on building an end-to-end analytics solution that integrates data cleaning, database design, SQL analysis, and interactive dashboards to provide a comprehensive view of business performance.

PROBLEM STATEMENT & OBJECTIVES

Problem Statement :

Despite steady revenue growth, RetailNova faces several challenges:

- Uneven regional and store-level performance
- Increasing product returns in certain categories
- Difficulty in identifying high-value customers
- Inefficient cost structures in some stores

Management requires a unified analytics solution to address these issues and support strategic decision-making.

Objectives :

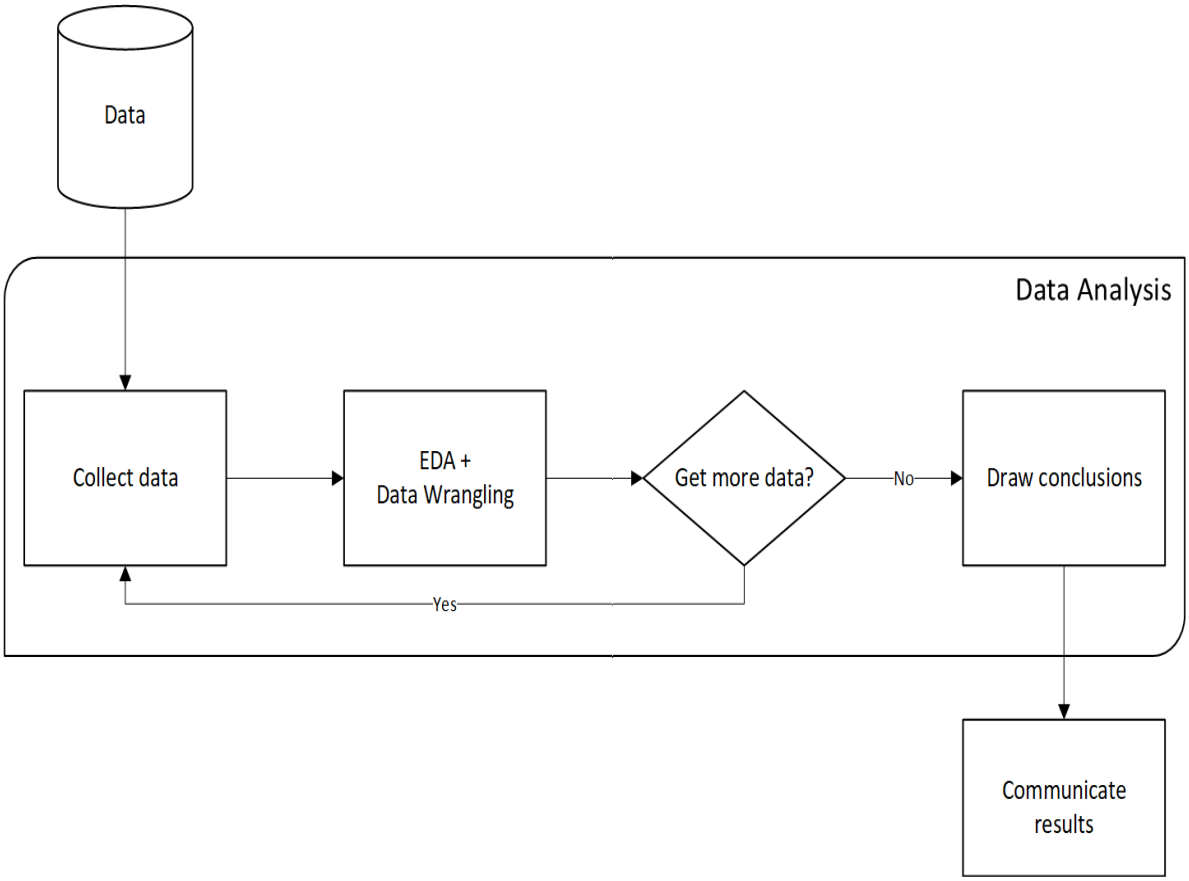
The objectives of this project are to:

- Analyse overall sales and revenue trends
- Understand customer demographics and behaviour
- Evaluate product profitability and return behaviour
- Assess store performance and operational efficiency
- Present insights through interactive dashboards

METHODOLOGY OVERVIEW

This project follows a structured analytics workflow:

1. Data Collection
2. Data Cleaning & Preprocessing
3. Feature Engineering
4. Exploratory Data Analysis
5. Database Design & SQL Analysis
6. Data Visualization & Reporting



DATA DESCRIPTION

The analysis uses five core datasets:

Dataset	Description
Sales	Transaction-level sales data
Customers	Customer demographics and signup details
Products	Product attributes and pricing
Stores	Store location and operating cost
Returns	Returned orders and reasons

Each dataset contributes to a different analytical dimension, enabling a holistic understanding of retail performance.

DATA CLEANING & FEATURE ENGINEERING

Data cleaning was performed using Microsoft Excel, followed by validation using Python and SQL. Key steps included:

- Removing duplicates
- Standardizing column formats
- Handling missing values
- Validating numerical fields

A dummy store record labelled **"Unknown"** was inserted to preserve referential integrity during SQL joins. This record was excluded at the visualization level to avoid analytical distortion.

Feature engineering included:

- Profit calculation
- Customer age grouping
- Rolling 12-month metrics

EXPLORATORY DATA ANALYSIS (EDA)

EDA was conducted using Python libraries such as Pandas and Matplotlib to understand data distributions and validate assumptions.

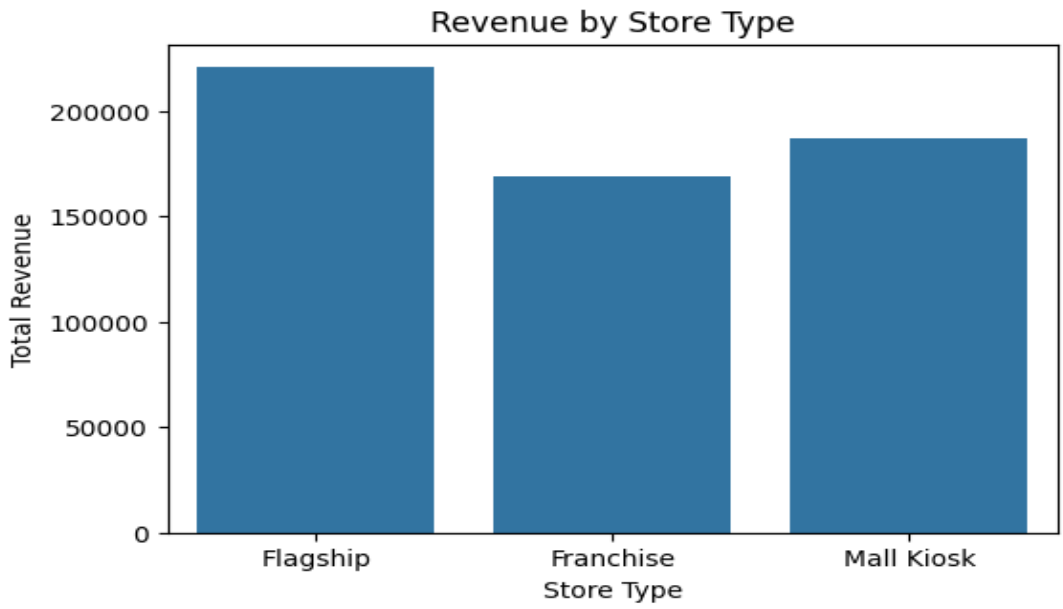


Fig: Revenue By Store Type

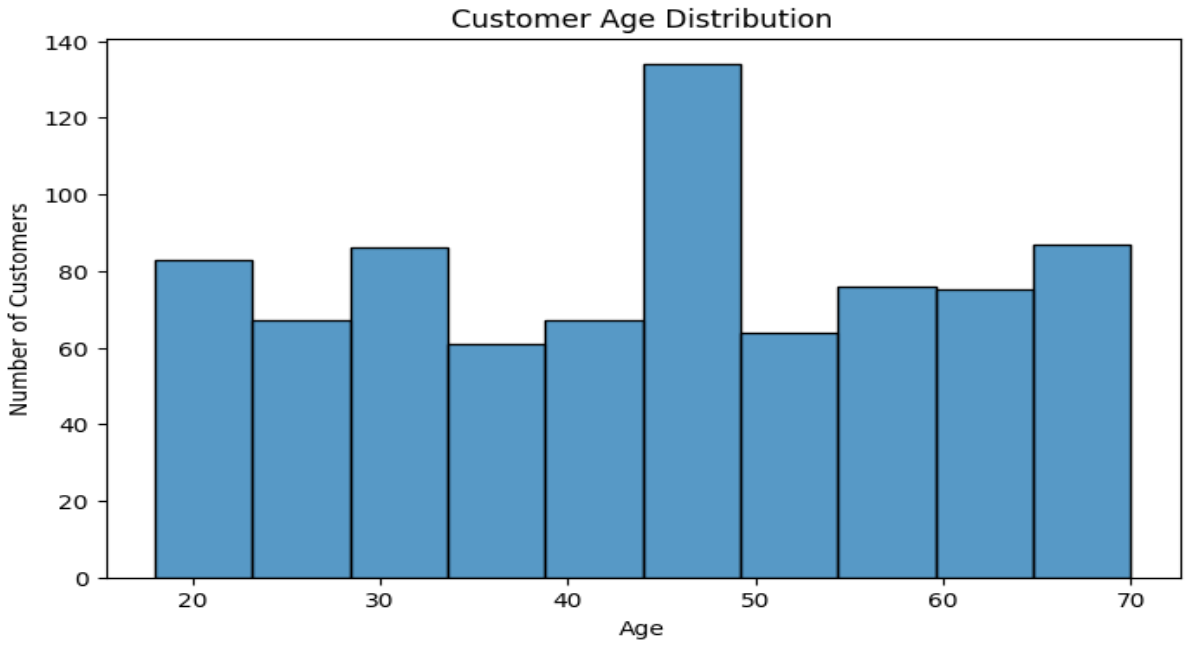
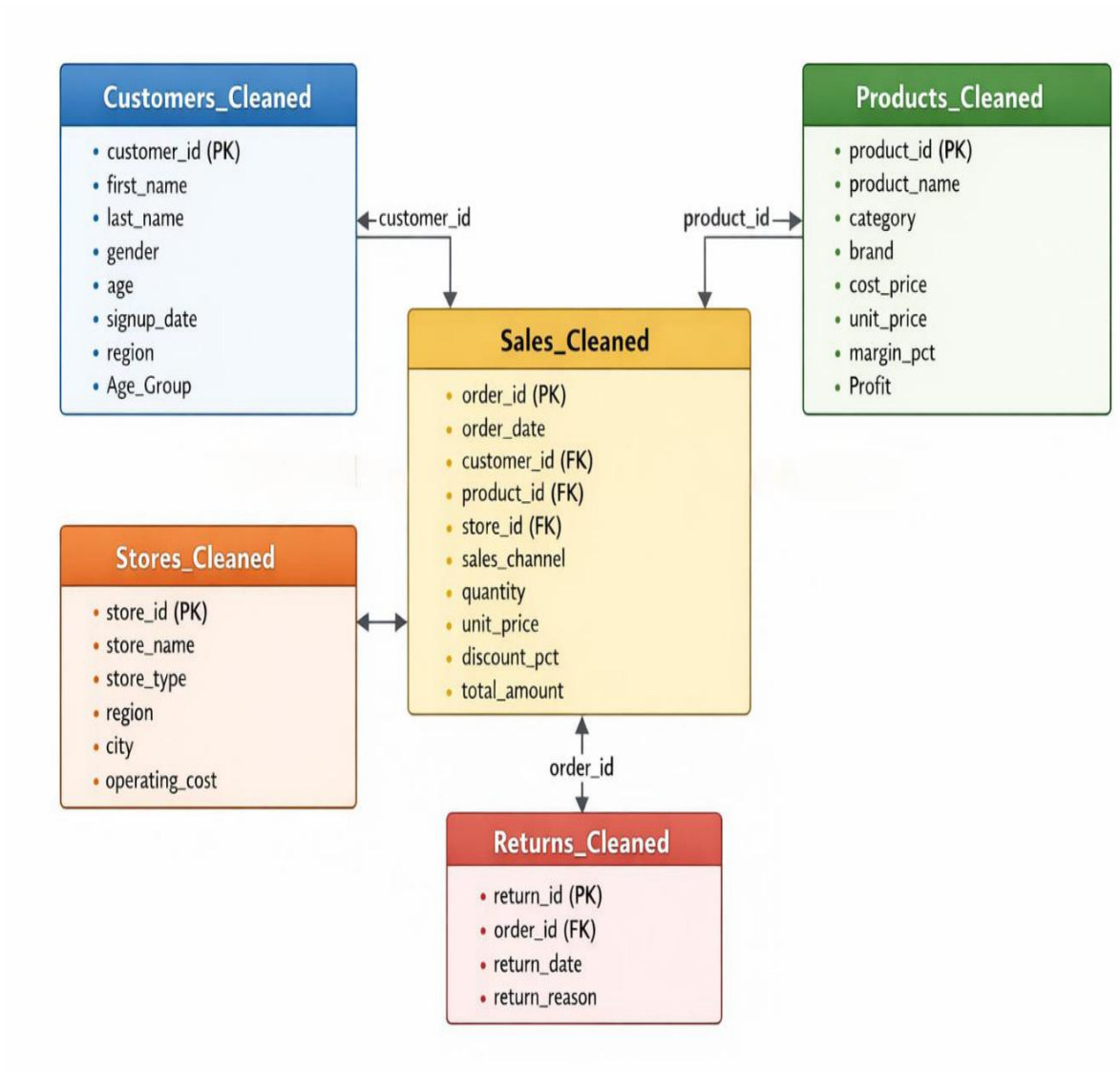


Fig: Customer Age Distribution

EDA confirmed data consistency and guided SQL query design.

DATABASE DESIGN

A relational database was implemented in Microsoft SQL Server using a fact–dimension structure. The **Sales_Cleaned** table acts as the central fact table, connected to customer, product, store, and return tables via foreign keys.



SQL ANALYSIS OVERVIEW

SQL was used to:

- Enforce relational integrity
- Compute derived metrics (revenue, profit, return rate)
- Answer business-driven analytical questions

Queries were modular, well-documented, and grouped logically for maintainability.

BUSINESS QUESTIONS ADDRESSED :

1. What is the total revenue generated in the last 12 months?

2. Which are the top 5 best-selling products by quantity?

3. How many customers are from each region?

4. Which store has the highest profit in the past year?

5. What is the return rate by product category?

6. What is the average revenue per customer by age group?

7. Which sales channel (Online vs In-Store) is more profitable on average?

8. How has monthly profit changed over the last 2 years by region?

9. Identify the top 3 products with the highest return rate in each category.

10. Which 5 customers have contributed the most to total profit, and what is their tenure with the company?

DASHBOARD OVERVIEW

Power BI dashboards were designed to provide interactive, self-service analytics across five perspectives.

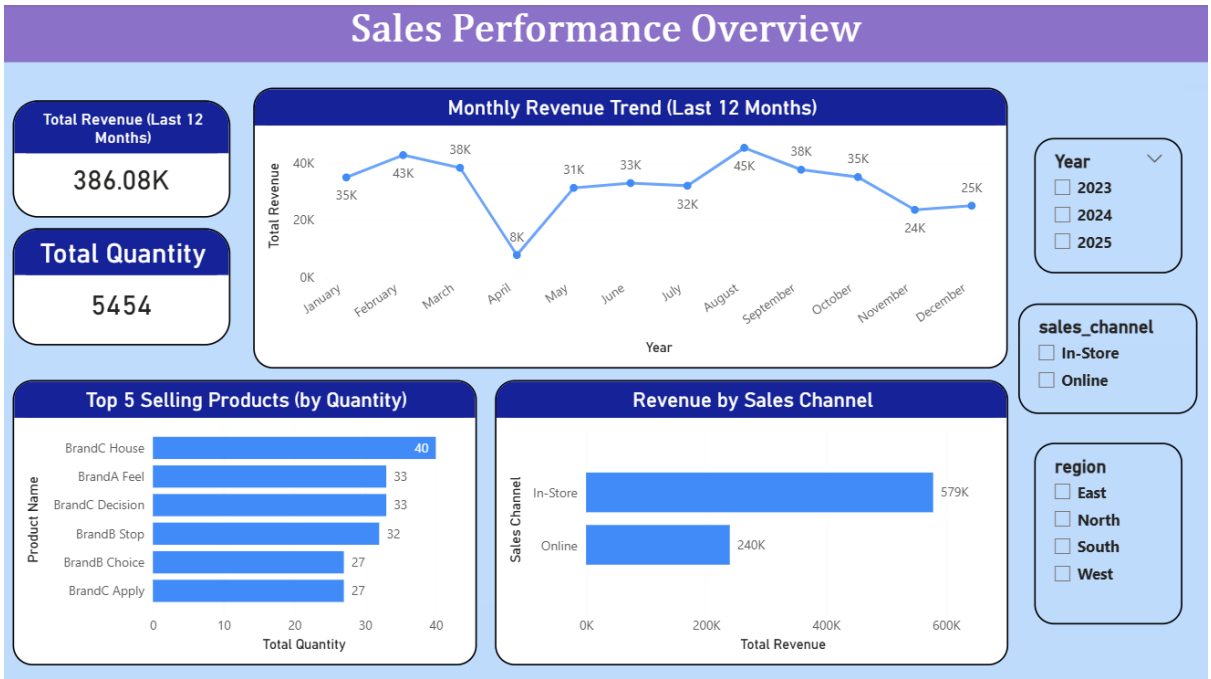


Fig: Sales Overview Report

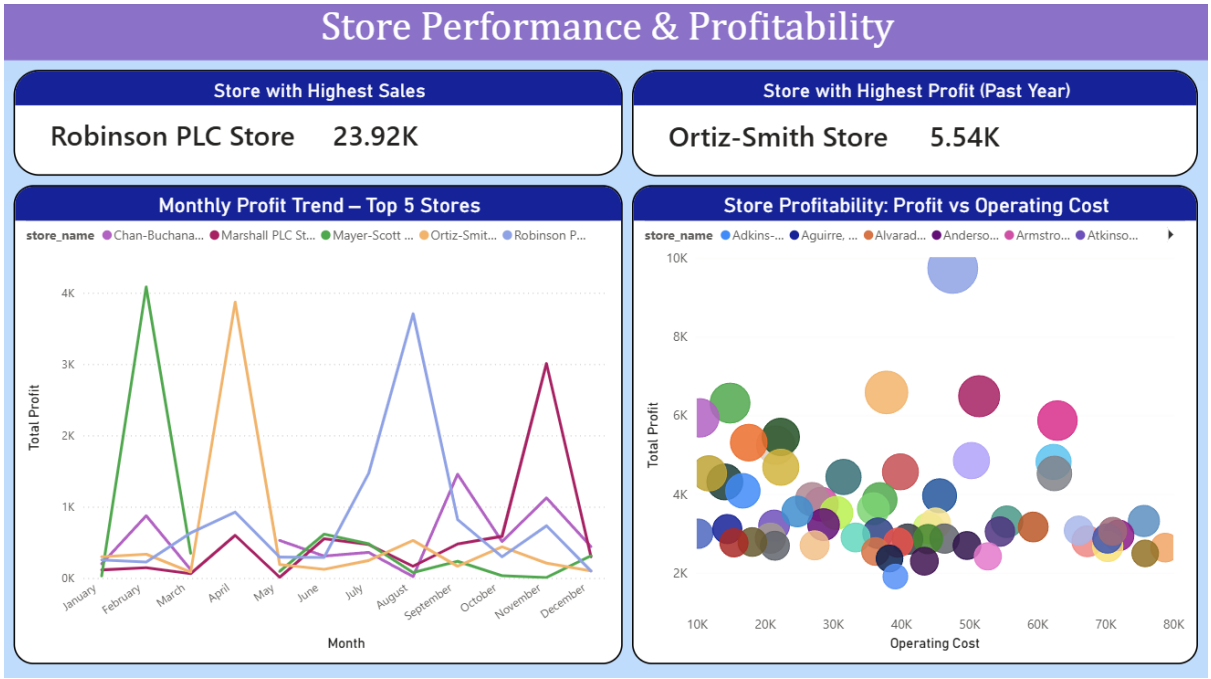


Fig: Store Analysis Report

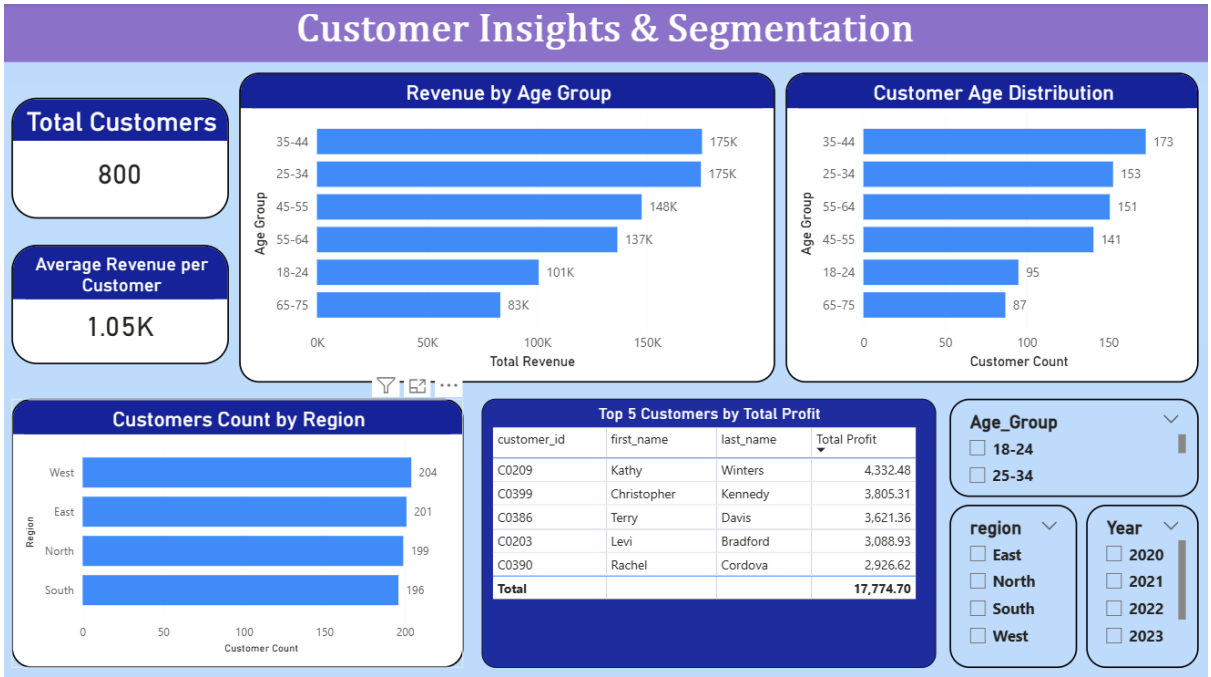


Fig: Customer Insights Report

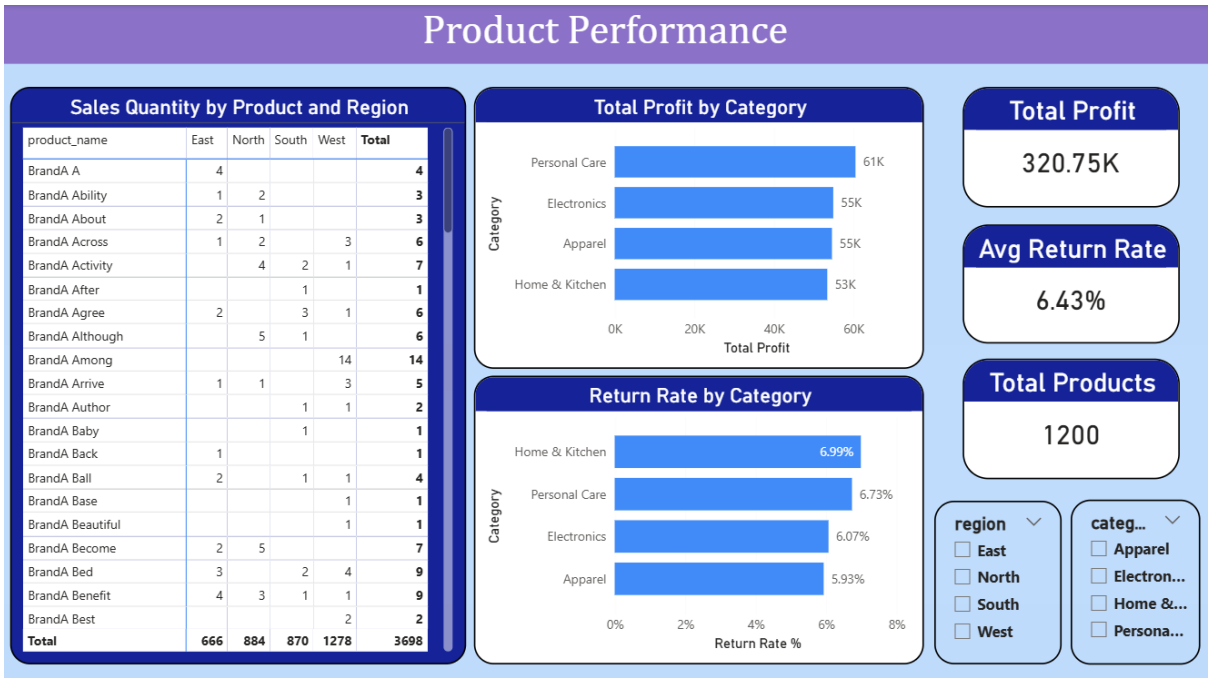


Fig: Product Performance Report

Fig: Return Analysis Report

KEY FINDINGS :

- Revenue shows consistent growth with seasonal variations
- In-store sales outperform online sales
- A small number of stores drive most profits
- Certain products contribute disproportionately to returns

BUSINESS INSIGHTS & RECOMMENDATIONS :

- Focus marketing on high-value age groups
- Optimize underperforming stores
- Investigate high-return products
- Improve online channel performance

CHALLENGES & LIMITATIONS :

- Limited behavioural data
- Low return volumes
- Absence of external factors

CONCLUSION

The dashboard helped to summarize key patterns such as top performing categories, customer trends, and monthly variations. The visualizations made it easier to identify which areas are strong and where improvements can be made. Overall, the reports provided a meaningful overview of the dataset and supported data-driven decision making in a simple and effective way.

REFERENCES

- Data Sources: customers, sales, products, return, store csv files.
- Tools Used: Python (Jupyter Notebook), SQL, Power BI.